MAPS4U STREET MAP SYSTEM (maps4u-maps) - COMPREHENSIVE TESTING REC

Release: v1.0.0-mvp | Gates G1-G6: ALL PASSED | Tests: 61/61 Passed | Coverage: 92% (2,412 stmts)

1. EXECUTIVE SMOKE & VERIFICATION SUMMARY

Suite / Check	Command	Result	Key Verified Metrics
End-to-End Smoke Test	python scripts/smoke_test.py	PASS	status=proof_ready, cache_hit=tr
Byte & Content Determinism	python scripts/verify_determinism.py	PASS	deterministic=true, content_hash
Hot SQLite + Bundle Drill	python scripts/backup_restore_drill.py	PASS	restored_ok=true, manifest_sha25
Static Linter (Ruff)	ruff check src tests scripts	PASS	All checks passed! (0 errors, 0
Strict Type Checker (Mypy)	mypy src	PASS	Success: no issues found in 46 s
Full Offline Pytest + Coverage	pytest -v --cov=src/maps4u	PASS	61 passed, 0 failed in 22.65s; 9

2. STAGE GATE PROGRESSION (G1 -> G6)

Gate	Tag	Scope & Deliverables Verified	Tests
G1	v0.1.0-phase0	Stage 1: FastAPI/Celery scaffold, SQLite WAL, Overpass fetch + 8-layer GPKG	14 passed
G2	v0.2.0-stage2	Stage 2: 6 paper templates (A3/A4/A5), 1:N scale & 1.5mm font viability, UI	42 passed
G3	v0.3.0-stage3	Stage 3: CairoCanvas PDF 1.4 vector renderer (ADR-0001), mcy/v1 YAML style	46 passed
G4	v0.4.0-stage4	Stage 4: STRtree label collision engine, map furniture, 2-page inset, QA bundle	54 passed
G5	v0.5.0-stage5	Stage 5: Full 11-stage job executor, review/approve lock, reproduce(), 3 pilots	58 passed
G6	v1.0.0-mvp	Stage 6: Multi-stage Dockerfile + compose, runbook, backup drill, full UAT	61 passed

3. THREE-PILOT CARTOGRAPHIC & REPRODUCIBILITY ACCEPTANCE (S5.T3)

Pilot Area	Format	Scale Band	Blockers	Overlaps/Clips	Min Font
Maroochydore (MCY Reference)	A4-landscape	standard_street (1:12500)	0	0 / 0	>= 1.50 mm
Noosa	A3-landscape	detail (1:7500)	0	0 / 0	>= 1.50 mm
Bribie Island - Sandstone Point	A4-portrait	standard_street (1:12500)	0	0 / 0	>= 1.50 mm

4. SMOKE TEST & DETERMINISM JSON OUTPUTS

smoke_test.py: {"smoke_test": "PASSED", "final_job_status": "proof_ready", "cache_hit_verified": true,
"job_id": "maroochydore-smoke-2026-10-smoke1-c19322", "canvas_hash": "73726d3c2c7f7c93af4d14a88a613898..."}

verify_determinism.py: {"deterministic": true, "content_hash": "46ec32956f277efb83104b4c1b5492c93b2b099019186cdc71584a1a59
"pdf_sha256": "dbc189d129cd34366ca5addeb474a339dbf014e916ddd8a678b2892e219b9b17"}

backup_restore_drill.py: {"restored_ok": true, "manifest_sha256_match": true, "pdf_sha256_match": true}

```
  /a0/usr/workdir/maps4u-street-map-system/src/maps4u/render/style_engine.py:184: DeprecationWarning: The 'shape
  /a0/usr/workdir/maps4u-street-map-system/src/maps4u/labels/engine.py:131: DeprecationWarning: The 'shapely.ops
  /a0/usr/workdir/maps4u-street-map-system/src/maps4u/labels/engine.py:173: DeprecationWarning: The 'shapely.ops
  /a0/usr/workdir/maps4u-street-map-system/src/maps4u/labels/engine.py:194: DeprecationWarning: The 'shapely.ops
  /a0/usr/workdir/maps4u-street-map-system/src/maps4u/labels/engine.py:218: DeprecationWarning: The 'shapely.ops
  /a0/usr/workdir/maps4u-street-map-system/src/maps4u/labels/engine.py:243: DeprecationWarning: The 'shapely.ops
tests/integration/test_stage6_packaging_and_handover.py::test_s6_t3_full_uat_operator_journey
tests/integration/test_stage6_packaging_and_handover.py::test_s6_t3_full_uat_operator_journey
tests/integration/test_stage6_packaging_and_handover.py::test_s6_t3_full_uat_operator_journey
Name                                    Stmts   Miss  Cover   Missing
---------------------------------------------------------------------
src/maps4u/__init__.py                      1      0   100%
src/maps4u/api/__init__.py                  0      0   100%
src/maps4u/api/deps.py                      6      0   100%
src/maps4u/api/main.py                     45      0   100%
src/maps4u/api/routes_jobs.py             124     15    88%   76, 83, 177-188, 199, 219, 226, 249, 334-344, 356
src/maps4u/api/routes_projects.py          59      6    90%   55-57, 79, 129, 171
src/maps4u/api/routes_styles.py             8      2    75%   14-15
src/maps4u/api/routes_system.py            20      0   100%
src/maps4u/api/routes_viability.py         23      2    91%   22-23
src/maps4u/bundle/__init__.py               0      0   100%
src/maps4u/bundle/archive.py               68      2    97%   95, 106
src/maps4u/core/__init__.py                 0      0   100%
src/maps4u/core/config.py                  51      0   100%
src/maps4u/core/db.py                      42      1    98%   52
src/maps4u/core/errors.py                  21      0   100%
src/maps4u/core/ids.py                     27      3    89%   35, 43-44
src/maps4u/core/logging.py                 30      6    80%   17-27
src/maps4u/core/models.py                  40      0   100%
src/maps4u/core/schemas.py                 79      0   100%
src/maps4u/core/state.py                   38      2    95%   62, 76
src/maps4u/geo/__init__.py                  0      0   100%
src/maps4u/geo/acquire.py                  30      0   100%
src/maps4u/geo/cache.py                    47      4    91%   25, 42-43, 85
src/maps4u/geo/corrections.py              98     12    88%   42, 45, 54, 88, 130, 134, 148-154
src/maps4u/geo/crs.py                      61     11    82%   31-32, 40, 53, 57-58, 61, 73, 82, 88, 95
src/maps4u/geo/features.py                 49     11    78%   94-104
src/maps4u/geo/fetch.py                    87      5    94%   47, 57, 125, 231-233
src/maps4u/geo/overpass.py                 44      5    89%   46-49, 74
src/maps4u/labels/__init__.py               0      0   100%
src/maps4u/labels/engine.py               210     24    89%   70, 72, 172, 175, 193, 196, 213, 217, 220, 238, 241,
src/maps4u/layout/__init__.py               0      0   100%
src/maps4u/layout/furniture.py            110      1    99%   123
src/maps4u/layout/templates.py            130     11    92%   48, 92, 116, 140, 152, 160, 164, 171, 199, 242, 248
src/maps4u/qa/__init__.py                   0      0   100%
src/maps4u/qa/preflight.py                 87      7    92%   65, 74, 83, 92, 101, 113, 172
src/maps4u/render/__init__.py               0      0   100%
src/maps4u/render/assess.py                85      3    96%   112, 125, 147
src/maps4u/render/cairo_canvas.py         210     14    93%   23, 25, 35, 37, 125, 138-139, 163, 189, 193, 219, 31
src/maps4u/render/canvas.py                59      7    88%   84-88, 91-92
src/maps4u/render/proof_renderer.py        64      0   100%
src/maps4u/render/style_engine.py         201     22    89%   79, 81, 85-86, 89, 128, 150, 161-162, 165-167, 222,
src/maps4u/worker/__init__.py               0      0   100%
src/maps4u/worker/celery_app.py             8      8     0%   3-25
src/maps4u/worker/executor.py             111      3    97%   35, 167, 169
src/maps4u/worker/pipeline.py              30      4    87%   53-67
src/maps4u/worker/tasks.py                  9      9     0%   3-16
---------------------------------------------------------------------
TOTAL                                    2412    200    92%
====================== 61 passed, 321 warnings in 22.65s =======================
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MAPS4U v1.0.0-mvp - PYTEST EXECUTION & COVERAGE RECORD (PAGE 3 OF 3)

61 Automated Tests (Unit, Property, Provider Contract, API, Golden Layout, PDF Structure, E2E UAT) + Coverage